Core Concepts in Quantum Field Theory

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This note is a summary of formal core concepts in quantum field theory. Including Symmetry, Renormalization, Anomalies, Solitons and Instantons. The note is mainly based on Peskin and Schroeder's "An Introduction to Quantum Field Theory"[1], Tong's lecture notes on gauge theory, and some other references.

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Contents

CHAPTER 1

Fundamental Ideas

This chapter serves as an introduction to basic ideas in QFT, which will be intensively used and discussed in the following chapters. It will not be self contained, yet all preliminaries(free field, path integral, Feymann diagrams) will be found in basic QFT textbooks.

1 LSZ Reduction Formula

For a scattering theory, it is important predict the S-matrix as observable quantity. For QFT, LSZ reduction formula is the one that tells us how to **calculate the S-matrix from the correlation functions**. Then we can:

1. Use Feymann diagrams to calculate correlation functions (Both functional derivative method or canonical quantization with interaction picture).
2. Use LSZ reduction formula to get the S-matrix from the correlation functions.

1.1 Kallen-Lehmann Form of Propagator

This is a topic to prepare us for the LSZ reduction formula. The question is:

- What is a “particle propagating” in a general Lorentz invariant QFT?

To have a concrete understanding of **particle** in interacting theory, we then have the tools of using QFT to describe the scattering of particles, and the S-matrix.

In free QFT, we know that the 2 point correlation function is interpreted as the amplitude of a particle propagating from one point to another:

$$\langle 0|\mathcal{T}\varphi(x)\varphi(0)|0\rangle \tag{1}$$

However, in general QFT, one may not expect this function to have such interpretation, since the propergator may include propergation of multiparticle states or even particle coming vaccum due to the interaction.

Thus, the role of the KL form is to tell us what part of the 2 point correlation function can be interpreted as the amplitude of a particle propagating, and what part is not.

1.1.1 Kallen-Lehmann Form of Scalar Propagator

First we have a look at the scalar propagator. We then need to specify our assumptions on the theory, which are:

1. The dynamical fields are scalar fields, and the theory is relativistic and Lorentz invariant.

2. The theory is a sensible quantum theory.

With these minimal assumptions, we can think of the spectrum of the theory can be organized as follow.

1.1.1.1 Spectrum of a Lorentz Invariant QFT

First, the Lorentz invariance of quantum theory means that the theory have a symmetry algebra of Poicare Algebra and the Hilbert space forms a representation of the Poincare Algebra.

A maximally commuting subalgebra of the momentum operator H, P^i, which can be used to label states in the theory. We consider a series of states in this theory with 0 eigenvalues of P^i. We will call them:

$$|\lambda_0\rangle \text{ satisfying } P^i|\lambda_0\rangle = 0 \quad H|\lambda_0\rangle = m_\lambda|\lambda_0\rangle \tag{2}$$

Then we can construct a series of states with nonzero eigenvalues of P^i by applying the Lorentz boost operator K^i on the states $|\lambda_0\rangle$, which will give us a series of states $|\lambda_p\rangle$:

$$|\lambda_p\rangle = e^{iK^i\xi_i}|\lambda_0\rangle \tag{3}$$

Due to the Poincare Algebra, we know that:

$$P^i|\lambda_p\rangle = p^i|\lambda_p\rangle \quad H|\lambda_p\rangle = \sqrt{p^2 + m_\lambda^2}|\lambda_p\rangle \tag{4}$$

Thus in general Hilbert Space of this Lorentz Invariant QFT take form of:

$$\mathscr{H} = \bigoplus_{\lambda,p}|\lambda_p\rangle \tag{5}$$

> **Remark**
>
> Notice that $|\lambda_p\rangle$ may not be a single particle states, but just a general momentum and energy eigenstate may have many particle and messy interaction and bounding between them.

1.1.1.2 Spectral Representation of Scalar Propagator

Consider this theory with the above spectrum. We can then write the 2 point correlation function as:

$$\langle 0|\mathcal{T}\varphi(x)\varphi(0)|0\rangle \tag{6}$$

Now we insert a complete set of states in the middle, which is:

$$\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum\!\!\!\!\!\!\int_\lambda \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p(\lambda)}|\lambda_p\rangle\langle\lambda_p| \tag{7}$$

> **Remark**
>
> notice that here we assume the relativistic normalization of states.

Notice that:

$$\langle\Omega|\varphi(x)|\Omega\rangle = 0 \tag{8}$$

This is due to the $\mathbb{Z}_2$ symmetry of the scalar theory, if we redefine $\varphi \to -\varphi$ the lagrangian is invariant, yet the correlation function will change sign, thus it must be zero. (For highier spin, the fact of being an irreducible representation of lorentz group will also lead to the vanishing of such one point function.)

Then we have:

$$\langle 0|\mathcal{T}\varphi(x)\varphi(0)|0\rangle = \sum\!\!\!\!\!\!\!\int_\lambda \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p(\lambda)}e^{-ipx}\langle\Omega|\varphi(x)|\lambda_p\rangle\langle\lambda_p|\varphi(0)|\Omega\rangle \tag{9}$$

Now have a look at the matrix element $\langle\Omega|\varphi(x)|\lambda_p\rangle$. We can translate the field to 0 by the property of $\varphi(x)$ being a lorentz scalar:

$$\langle\Omega|\varphi(x)|\lambda_p\rangle = \langle\Omega|e^{iPx}\varphi(0)e^{-iPx}|\lambda_p\rangle = e^{-ipx}\langle\Omega|\varphi(0)|\lambda_p\rangle \tag{10}$$

By being a scalar, it must satisfy:

$$U(\Lambda)\varphi(0)U^\dagger(\Lambda) = \varphi(0) \tag{11}$$

Thus we have:

$$\langle\Omega|\varphi(0)|\lambda_p\rangle = \langle\Omega|U^\dagger(\Lambda)U(\Lambda)\varphi(0)U^\dagger(\Lambda)U(\Lambda)|\lambda_p\rangle = \langle\Omega|\varphi(0)|\lambda_0\rangle \tag{12}$$

1.1.1.3 Kallen-Lehmann Form of Scalar Propagator

1.1.2 Kallen-Lehmann Form of Fermion Propagator

1.1.3 Example: QED Fermion Propagator

Now we have a look at a concrete example of a lorentz invariant QFT, which is QED.

1.2 LSZ Reduction Formula

1.2.1 Derivation of LSZ Reduction Formula

1.2.2 S-Matrix from Feymann Diagrams

2 Ward-Takahashi Identity

Symmetry plans the fundamental role in QFT, and the Ward-Takahashi identity is a consequence of the symmetry. I'll first introducs the formal structure of the Ward-Takahashi identity following section 9.6 of [1], and then talk about the application in electrodynamics, following section 7.4 of [1].

2.1 Ward-Takahashi Identity in General QFT

3 Quantization of Gauge Fields

CHAPTER 2

Renormalization

Renormalization is the procedure to deal with divergences in quantum field theory. It is a systematic way to cancel the divergences and get finite results for physical quantities.

1 Regualization

Before starting with the systematic renormalization canceling out divergence, we need to first regularize the divergences, which is the process to make the divergences well defined. This is done by introducing a regularization scheme, which is a mathematical tool to make the divergences well defined.

1.1 Regularization Schemes

To regularize the divergences, we need to first calculate the loop integrals to some good extent, and the introduce the regularization scheme to make the divergences well defined. There are several regularization schemes, and the most common ones are:

- Pauli-Villars Regularization
- Dimensional Regularization

This section we will intensively study how to calculate loop integrals and do regularization in a standard way. Here are the general steps to folow:

1. Use **Feymann Parameters** to combine the denominators of the propagators in the loop integrals, which will make the loop integrals easier to calculate.
2. Make l

1.1.1 Feymann Parameterization

1.1.2 Dimensional Regularization

1.2 Examples

Lets use the mathematical tools developed in the previous sections to calculate some examples of loop integrals and do regularization, which will be useful for the following sections on renormalization.

1.2.1 Example: Electric Vertex Function

This is the most basic example of renormalization, which is the one-loop correction to the electric vertex function, given in section 6.3 and 7.2, 7.3 of [1].

1.2.2 Example: Vaccum Polarization

This is another example from Peskin and Schroeder [1], which is the one-loop correction to the photon propagator, given in section 7.5.

2 Renormalization

We have seen that the observables we care about seems to have divergences due to the loop integrals. Now we need to find a way to understand why there are these divergences and how to get finite results for physical quantities. This is the process of renormalization.

2.1 Bare Perturbation Theory

Before going to a systematic way to do renormalization, we need to first try to do it in a more naive and cumbersome way called **bare perturbation theory**. Yet it may help us to understand why the systematica **renormalized perturbation theory** works.

2.1.1 General Method

Before we have encountered divergences in observables, seems like the theory is ill defined. However, in fact its is our understanding of the theory is incomplete, here is a modern understanding:

- Before, we have infinity due to the fact that we are using the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$ to express the S Matrix, thus the coefficients turns out to be infinite.
- We have to define some physical quantities (and argue why they are physical) and think of them as the result of stuffs we really measure in experiments, such as the physical mass m, the physical coupling constant λ ...
- Now we can express the S Matrix in terms of the physical quantities, turns out that the coefficients are finite, and we can get finite results for physical quantities.

To do this, we follow the following steps:

1. **Regularize the divergences in Observable** (eg. S Matrix), and express the Observable in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$, with infinite divergences.
2. **Define the physical quantities**, such as the physical mass m, the physical coupling constant λ and calculate them perturbatively in terms of the unobservable infinite quantities $m_0, \lambda_0, \Lambda, Z, ...$ to some orders.
3. **Express the Observable in terms of the physical quantities**, and we will find that the coefficients are finite, and we can get finite results for physical quantities.

> **Note**
>
> It is important to have all steps in the same order of perturbation. For tree level, there are no loops and divergence, thus we can directly express bare quantity as physical quantities, if we fix the order of perturbation to be tree level.

2.1.2 Example: Electric Vertex Function

The first example is the one-loop correlation to the electric vertex function. We may see that though we have divergences in correlation functions. However, if we focus on physical quantities, such as S Matrix and use the physical mass instead of bare mass. The divergence will disappear.

2.1.2.1 Divergence in all orders

There is a problem of can field strength renormalization cancel divergence to all orders? this is enforced by the Ward-Takahashi identity, which is a consequence of the gauge symmetry.

2.1.3 Example: Vaccum Polarization

2.2 Renomalizability

Above sections we show that some of the divergence in a theory may be canceled out using a renomalizing technique. However, for a sensible theory, we need to make sure that all the divergences can be canceled out when expressing the physical obserables using physical values.

Thus, before developing a systematic way to do renormalization, we need to first classify theories based on whether they can be renomalized or not.

2.2.1 Which Theories are Renormalizable?

To answer this question, we have 2 subquestions:

1. How to count the number of divergent amplitudes in a theory?
2. What kind of divergence is controllable? (ie. can be canceled out by renormalization)

2.2.2 How to Count Divergences?

We have to answer the first question first.

- How to count the number of divergent amplitudes in a theory?

We introduce the concept of **superficial degree of divergence**. To make things easy, we take a φ^n theory in d dimensional spacetime as an example:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2}m^2\varphi^2 - \frac{\lambda}{n!}\varphi^n \tag{13}$$

Now consider a Feymann Diagram with:

$$\begin{aligned} &N : \text{number of external lines} \\ &P : \text{number of propagators} \\ &V : \text{number of vertices} \\ &L : \text{number of loops} \end{aligned} \tag{14}$$

We can define a quantiy called **superficial degree of divergence** as:

Definition 2.1 **Superficial Degree of Divergence**

The superficial degree of divergence of a Feymann Diagram is defined as:

$$D = dL - 2P \tag{15}$$

where L is the number of loops and P is the number of propagators in the Feymann Diagram.

Why does this quantity characterize the divergence of the Feymann Diagram? We can have a loop at the Feymann Rules:

- Naively, each Propagator contributes a factor of $1/k^2$ to the amplitude, which gives a factor of -2 to the degree of divergence.

$$D_F \sim \frac{1}{k^2 + m^2} \tag{16}$$

- Each loop contains a momentum that needs to be integrated over, which gives a factor of;

$$\int \frac{d^d k}{(2\pi)^d} \tag{17}$$

Thus the total contribution to the degree of divergence is:

$$D = dL - 2P \tag{18}$$

There is also an expression using numbers of vertices and external lines:

$$\begin{gathered} L = P - V + 1 \quad nV = 2P + N \\ \Rightarrow D = d - \frac{d-2}{2}N - V\left(d - n\frac{d-2}{2}\right) \end{gathered} \tag{19}$$

However, we know that this is really just a simplified version of the true divergence. Because there are some special cases that have different divergence than the superficial degree of divergence.

Commonly these special cases are given in the following list (this list is exactly true for QED, that this list contains all the special cases for QED):

1. **Sub-Diagram Divergence**: A Feymann Diagram may contain a sub-diagram that has a divergence, which may gives larger divergence than the superficial degree of divergence.
2. **Symmetry**: Some Feymann Diagrams may have some symmetry that ensures some divergent diagrams canceled out in some amplitudes, which makes the true divergence smaller than the superficial degree of divergence.
3. **Trivial diagram**: trivial diagram with no loops and no propagators, which has a degree of divergence of $D = 0$ but is actually finite.

We can argue that these special cases have no effect on our analysis of divergence and renormalizability:

1. For the first case, we can just consider all amputated, one particle irreducible basic amplitudes, and make sure they are controlable. If one highier diagram is superficially converge, yet truely diverge, the only possibility is to have a sub-diagram in the cases we discussed, which is controlable.
2. For the second case, we would like to see this.
3. For the third case, just ignore it, no one cares about it.

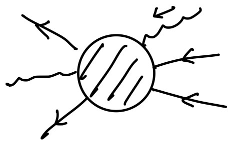
Thus, we'd like to say that the superficial degree of divergence is a good way to characterize the divergence of the Feymann Diagram. There are conter examples, especially for more complicated theories like Yang-Mills theory, this will be discussed in more advanced sections.

2.2.2.1 What kind of Divergence is Controllable?

The second question is what kind of divergence is controllable, which means that we can cancel out the divergence by renormalization. The answer is that:

- **Super Renormalizable**: Theories with only a finite number of superficially divergent diagrams: which means that 1. only a finite number of amplitudes diverges 2. they only diverge to a finite order.

- **Renormalizable**: Theories with an infinite number of superficial divergent amplitudes: which means that 1. an infinite number of amplitudes diverges 2. they may diverge in any order.
- **Nonrenormalizable**: Theories with an infinite number of superficial divergent amplitudes.

> **Remark**
>
> Here for Amplitude, we means general stuffs like this, which contains summing over all Feymann Diagrams to all orders.
>
> 
>
> If an amplitude diverges, it may have two cases: either Feymann Diagrams to all orders have divergence or only a Feymann Diagram of some finite order diverges. This difference separates the superrenormalizable theories and renormalizable theories.

As the name suggests, only a renormalizable theory is renormalizable.

- Why is this the case??

A basic analysis is that ther are only finite number of parameters we can use as physical quanitities. If there are too many infinities, they may not be adiquate enough for cancelation. And if we have finite numbers of amplitudes that diverges superficially, we can have finite physical quantities to plug in and cancel out the divergences. We will see examples to justify this claim.

Thus we can have a partically systematic way to see whether a theory is renormalizable or not:

- **Step 1:** try to list out all superficially divergent, amputated, one particle irreducible basic amplitudes.
- **Step 2:** check whether there are truely divergent or not; there are more divergent amplitudes beyond superficial expectations.

We will finally see examples of this procedure.

2.2.3 Mass Dimension and Renormalizability

2.2.3.1 Divergence and Mass Dimension

There is also a more straightforward way of seeing divergence of a Feymann Diagram. We know that an amplitude should be of the same dimension.

consider a n external leg diagram in the φ^n theory:

- The tree level diagram:

$$i\mathcal{M} = -i\lambda \tag{20}$$

By dimensional analysis, we have:

$$[\lambda] = d - n\frac{d-2}{2} \tag{21}$$

- Consider a loop diagram of this n external leg amplitude, which has V vertices, the amplitude is:

$$i\mathcal{M} \sim \lambda^V \Lambda^D \tag{22}$$

where Λ is the cutoff scale, and D is the superficial degree of divergence.

- By dimensional analysis, we have:

$$[\lambda^V \Lambda^D] = V\left(d - n\frac{d-2}{2}\right) + D = d - n\frac{d-2}{2} \tag{23}$$

inverse the equation, we have:

$$D = d - n\frac{d-2}{2} - V\left(d - n\frac{d-2}{2}\right) \tag{24}$$

which is exactly the same as the superficial degree of divergence we have defined before.

2.2.3.2 Renormalizability from Mass Dimension

These relations in fact is much physical than just a mathematical relation. We then can see the mass dimension of the coupling constant gives us a direct way to see whether a theory is renormalizable or not (of course, superficially):

- **Super Renormalizable**: Theories with coupling constants of positive mass dimension.
- **Renormalizable**: Theories with coupling constants of zero mass dimension.
- **Non Renormalizable**: Theories with coupling constants of negative mass dimension.

In fact this can also be justified in a EFT view.

YL: [Discussion to be continued, for the Wilsonian view of QFT and their consistency with the renormalization procedure.]

2.2.4 Renormalizability of QED

Lets apply this two steps to QED:

- **Step 1:** try to list out all superficially divergent, amputated, one particle irreducible basic amplitudes.
- **Step 2:** check whether there are truely divergent or not; there are more divergent amplitudes beyond superficial expectations.

The list of all superficially divergent, amputated, one particle irreducible basic amplitudes in QED is given in the following table:

2.3 Renormalized Perturbation Theory

We have seen that the bare renormalization is cumbersome and may have some potential problems unclear in the procedure.

2.3.1 General Method

Renormalization Condition is in fact defining what the physical quantities are in this theory.

2.3.2 Understanding From LSZ

Its good to mention [2]'s understanding of renormalization and counter terms from interpreting the LSZ reduction formula.

2.4 Examples of RPT

2.4.1 Example: φ^4 Theory

CHAPTER 3

Renormalization Group

Before we deal with

CHAPTER 4

Spontaneous Symmetry Breaking

The vaccum state may in general have different symmetries than the Lagrangian. This is a well know result in many cases. In QFT, this is know as the spontanous symmetry breaking. This phenomenon is physical in real life systems and also have important physical consequences for renomalization.

CHAPTER 5

Anomalies

1 Chiral Anomalies

This section I will follow David Tong's lecture on gauge theory to talk about chiral anomalies in fermions when coupled to gauge fields.

2 Gauge Anomalies

3 't Hooft Anomaly

Bibliography

[1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory.* Reading, Mass: Addison-Wesley Pub. Co, 1995.

[2] M. A. Srednicki, *Quantum Field Theory.* Cambridge ; New York: Cambridge University Press, 2007.